Sec: SR.IIT_*COSC(MODEL-A&APEX) | GTM-11 | Date: 20-03-26
Time: 3 Hrs | JEE-MAIN | Max. Marks: 300

KEY SHEET

PHYSICS

1	**C**	2	**A**	3	**D**	4	**C**	5	**D**
6	**B**	7	**D**	8	**A**	9	**A**	10	**A**
11	**D**	12	**A**	13	**A**	14	**B**	15	**D**
16	**D**	17	**C**	18	**A**	19	**C**	20	**A**
21	**4**	22	**3**	23	**885**	24	**2**	25	**10449**

CHEMISTRY

26	**A**	27	**C**	28	**A**	29	**D**	30	**C**
31	**A**	32	**B**	33	**B**	34	**B**	35	**B**
36	**D**	37	**C**	38	**C**	39	**B**	40	**A**
41	**C**	42	**A**	43	**D**	44	**C**	45	**D**
46	**2**	47	**3**	48	**5**	49	**3**	50	**3**

MATHEMATICS

51	**B**	52	**A**	53	**C**	54	**D**	55	**D**
56	**B**	57	**A**	58	**B**	59	**B**	60	**A**
61	**B**	62	**C**	63	**B**	64	**A**	65	**B**
66	**D**	67	**B**	68	**A**	69	**A**	70	**A**
71	**280**	72	**95**	73	**9**	74	**6**	75	**218**

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SOLUTIONS

PHYSICS

1. $F = K_1x_1, x_1 = \frac{F}{K_1}, W_1 = \frac{1}{2}K_1x_1^2 = \frac{F^2}{2K_1}$

2. $a = g\sin\theta - (\mu_0 g\cos\theta)x \;\; \therefore \int_0^v vdv$

$= \int_0^{x_{max}} [g\sin\theta - (\mu_0 g\cos\theta)x]dx$

After solving we get $x_{max} = \frac{2\tan\theta}{\mu_0}$

3. $P_{ar} = \frac{1}{2}\mu v\omega^2A^2$

So $P \propto v$

$\propto \omega^2$

$\propto A^2$

So if frequency or amplitude is doubled power will be quadrupled, but if velocity is double power will also be doubled.

4. As $U(x) = U_0(1 - \cos a\,x)$

So, $F_x = -\frac{dU}{dx} = -U_0 a\sin a\,x$ (1)

Or $F_x = -U_0$ a ax(because for small angle of oscillations sin ax $\equiv$ ax)

Or $F_x = -U_0a^2x$ (1)

But we know $F_x = -m\omega_0^2x$, for small oscillation

Thus $\omega_0^2 = \frac{U_a a^2}{m}$ or $\omega_0 = a\sqrt{\frac{U_0}{m}}$

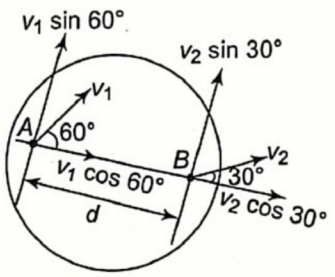
Hence the sought time period

$T = \frac{2\pi}{\omega_0} = \frac{2\pi}{a}\sqrt{\frac{m}{U_0}} = 2\pi\sqrt{\frac{m}{a^2U_0}}$

5. For rigid body, separation between two points remains same.

$v_1\cos 60° = v_2\cos 30°$

$\frac{v_1}{2} = \frac{\sqrt{3}v_2}{2} \Rightarrow v_1 = \sqrt{3}v_2$



$\omega_{disc}\left|\frac{v_2\sin 30° - v_1\sin 60°}{d}\right| = \frac{\left|\frac{v}{2} - \frac{\sqrt{3}v_1}{2}\right|}{d}$

$= \frac{|v_2 - \sqrt{3}\times\sqrt{3}v_2|}{2d} = \frac{2v_2}{2d} = \frac{v_2}{d} \Rightarrow \omega_{disc}$

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6. To find the value of x for which particles A and B collide, we need to set their position equations equal to each other at the same time. The position of particle A at time t is given by:

$\vec{r}_A(t) = \vec{r}_A + \vec{v}_A t = (2\hat{i} + 3\hat{j}) + (3\hat{i} - \hat{j})t$

The position of particle B at time t is given by:

$\vec{r}_B(t) = \vec{r}_B + \vec{v}_B t = (6\hat{i} + 7\hat{j}) + (x\hat{i} - 5\hat{j})t$

Setting these equal gives us two equations : 1.

For the i components

$2 + 3t = 6 + xt$

and 2. For the j components

$3 - t = 7 - 5t$

We will solve these equations to find the value

Step 1

From the j components, set up the equation:

$3 - t = 7 - 5t$

Step 2

Rearranging gives:

$5t - t = 7 - 3$

Which simplifies to

$4t = 4$

Thus

$t = 1$

step 3

Now substitute t = 1 into the i components equation:

$2 + 3(1) = 6 + x(1)$

Which simplifies to

$5 = 6 + x$

Step 4

Rearranging gives:

$x = 5 - 6 = -1$

Final Answer : -1

7. $F_1 = 0, F_2 = 0$

$a_{s_2 s_1} = a$ …(1)

Acceleration of the particle w.r.t. to $S_1 = F_1 / m$

Acceleration of the particle w.r.t. to $S_2 = F_2 / m$

If we assume $F_1 = 0$ and $F_2 = 0$, we can conclude that $a_{s_2 s_1} = 0$ …(2)

From equation (1) and (2), we can say that our assumption is wrong and $F_1 = 0, F_2 = 0$ is not possible.

8. Carnot engnine is an ideal engine so its efficiency will be maximum

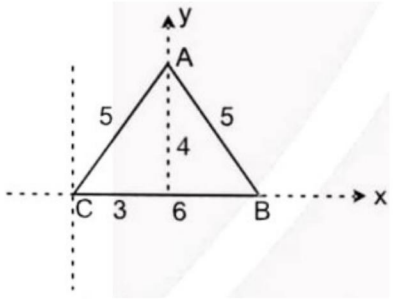
$\therefore \eta_{max} = \frac{400 - 300}{400} \times 100\% = 25\%$

Therefore 26% efficient engine is impossible

9. Increasing the temperature of water from 2°C to 3°C increases its density while decreases the density of iron. Hence the bouyant force increases.

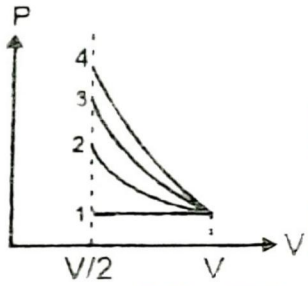
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10.



$y_{cm} = \frac{0\times(2\rho)+2\times5\rho\times2}{2(\rho)6+2\times5\times\rho} = \frac{10}{11}$ distance from $A = \left(4-\frac{10}{11}=\frac{34}{11}\right)$

11.



So, final pressure is maximum for PV^2 = constant.

12. $\frac{R^2}{8h}+2h = \frac{\left(\frac{u^2 2\sin\theta\cos\theta}{g}\right)^2}{\frac{8\times u^2\sin^2\theta}{2g}} + 2\frac{u^2\sin^2\theta}{2g} = \frac{u^2}{g}$ (max. horizontal Range)

13. When train A moves form east to west

$mg - N_1 = \frac{m(v+\omega R)^2}{R}$

$\Rightarrow N_1 = mg - \frac{m(v+\omega R)^2}{R}$

$N_1 = F_1$

When train B moves from west to east $mg - N_2 = \frac{m(v-\omega R)^2}{R} \Rightarrow N_2 = mg - \frac{m(v-\omega R)^2}{R}$

$N_2 = F_2$

$F_1 > F_2$

14. $C = \frac{k\varepsilon_0 A}{d}$

$Q = CV$

$F = \frac{q^2}{2\varepsilon_0 A}$

$U = \frac{1}{2}CV^2$

$E = \frac{V}{d}$

15. F_3 & F_4 cancel each other

Force on PQ will be $F_1 = I_2 B_1$ a

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$= I_2 \frac{\mu_0 I_1}{2\pi a} a$

$= \frac{\mu_0 I_1 I_2}{2\pi}$

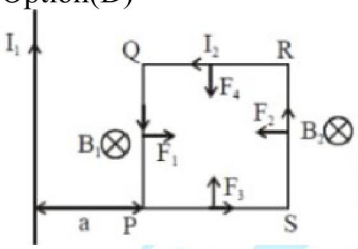
Force on RS will be $F_2 = I_2 B_2 \, a$

$= I_2 \frac{\mu_0 I_1}{1\pi 2a} a$

$= I_2 \frac{\mu_0 I_1}{1\pi 2a}$

Net force $= F_1 - F_2 = \frac{\mu_0 I_1 I_2}{4\pi}$ repulsion

Option(D)

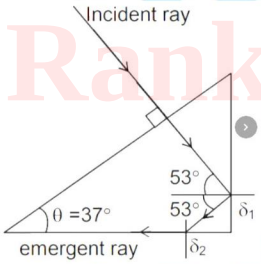


16. Diode D_1 is in reversed biased so current through it will not flow in this branch.

Equivalent resistance $= 150 + 100 = 250\Omega$

$I = \frac{12}{250} = \frac{48}{1000} = 48mA$

17. $\delta_1 = 180° - 2 \times 53° = 74°$



$\delta_1 = 53°$

$\delta_{net} = \delta_1 + \delta_2 = 127°$

18. $P = \frac{E_{rms}^2 \cos\phi}{z} \Rightarrow 400 = \frac{(250)^2}{z} \times 0.8$

$\cos\phi = \frac{R}{Z} = 0.8$

$z = \sqrt{X_L^2 + R^2}$

$R = 100\Omega$

$Z = 125\Omega$

$X_L = 75\Omega$

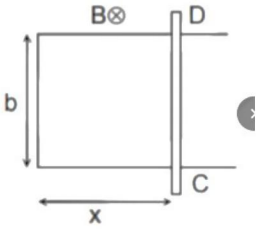
19. No. of photons per second $= \frac{1}{\left(\frac{hC}{\lambda}\right)} \times \text{Area} = \frac{8 \times 10 \times 10^{-4}}{2 \times 1.6 \times 10^{-19}}$

No. of photoelectrons per second $= \frac{1}{5} \times \frac{8 \times 10 \times 10^{-4}}{2 \times 1.6 \times 10^{-19}} = 5 \times 10^{15}$

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20. The magnetic flux must remain constant.

$$\therefore \phi_m = B_0 ab = \frac{B_0}{1+kt} bx$$



$$\therefore x = a(1+kt)$$

$$\text{Or } v = \frac{dx}{dt} = ak$$

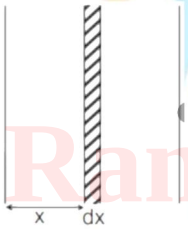
21. $V_A - 2 - 6 + 4 = V_B$

$$V_A - V_B = 4V$$

22. $$dC = \frac{kL^2\varepsilon_0}{dx}$$

$$dC = \frac{B}{x^2}\frac{L^2\varepsilon_0}{dx}$$

$$\frac{1}{C_e} = \int \frac{1}{dC} \Rightarrow C_e = \frac{3BL^2\varepsilon_0}{d^3}$$



23. $$l = \frac{1}{2}\varepsilon_0 E_0^2 c$$

$$\text{Radiation pressure} = \frac{2l}{c} = \varepsilon_0 E_0^2 = 8.85\times10^{-12}\times10^4 = 885\times10^{-10}\,\text{Pa}$$

24. $$\phi = -\frac{2\pi}{\lambda} = \frac{2\pi}{\lambda}\frac{\lambda}{4}$$

$$\phi = \frac{\pi}{2}$$

$$I_R = I_0\cos^2\left(\frac{\phi}{2}\right) = I_0\cos^2\frac{\pi}{4}$$

$$I_R = \frac{I_0}{2}$$

25. $$\lambda = \frac{c}{\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)}$$

for lyman series

$$\lambda_1 = \frac{c}{\frac{1}{1^2} - \frac{1}{\infty^2}} = c(n=\infty \text{ to } n=1)$$

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$$\lambda_2 = \frac{c}{\frac{1}{1^2}-\frac{1}{2^2}} = \frac{4c}{3}(n=2\,to\,n=1)$$

$$\Delta\lambda = \lambda_2 - \lambda_1 = \frac{c}{3} = 301\,\text{Å} \Rightarrow c = 903\,\text{Å}$$

For paschen series

$$\lambda_1 = \frac{c}{\frac{1}{3^2}-\frac{1}{4^2}} = 9c(n=\infty \text{ to } n=3)$$

$$\lambda_2 = \frac{c}{\frac{1}{3^2}-\frac{1}{4^2}} = \frac{144c}{7}(n=4 \text{ to } n=3)$$

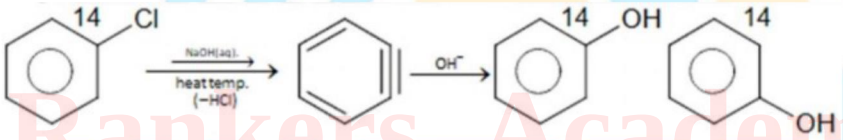
$$\Delta\lambda = \lambda_2 - \lambda_1 = \frac{144c}{7} - 9c = \frac{81c}{7} = \frac{81\times 903}{7}$$

$$= 10449\,\text{Å}$$

CHEMISTRY

26. (A) Lassaigne's test is used to identify the elements N,S,P and halogens
(B) Cu(II) O is used to identify 'Carbon'
(C) $AgNO_3$ is used to identify halogen specifically
(D) $2Na^+ + S^{-2} \rightarrow Na_2S \xrightarrow{CH_3COOH} H_2S \xrightarrow{pb(CH_3COO)_2} PbS\downarrow (black)$

27.



28. $C_6H_5N + H_2O \rightleftarrows C_6H_5NH_3^+ + OH^-$

1	0	0
$1-\alpha$	α	α

($\because$ Dissociation occurs in presence of NaOH and thus dissociation of $C_6H_5NH_2$ will suppress)

$$K_b = \frac{\left[C_6H_5NH_3^+\right]}{\left[C_6H_5NH_2\right]} = \frac{10^{-8}\times[OH^-]}{[0.24]}$$

$$\because K_b \text{for} C_6H_5NH_2 = \frac{K_w}{K_a \text{for}\, C_6H_5NH_3^+} = \frac{10^{-14}}{2.4\times10^{-5}}$$

$$\therefore [OH^-] = \frac{10^{-14}\times 0.24}{2.4\times10^{-5}\times10^{-8}} = 0.01$$

$\therefore NaOH = 10^{-2}M$

29. Purification process in chromatography doesn't depend on physical state of pure compound.
30. Refer ncert
31. Refer ncert
32. Compound II is non aromatic
33. PBr_5 exists as $\left[PBr_4^+\right]\left[PBr_6^-\right]$ in solid state
34. Presence of EWG – on diazonium ions at o & p-positions favour coupling reactions. Diazonium ion without electron withdrawing groups does not couple with anisole.
35. The velocity of electron is an orbit can be calculated by using the formula.

$$v = \frac{2.18\times10^6\times z}{n} m/sec$$

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$\therefore$ 'v' is directly proportional to 'z' and is inversely proportional to 'n' $\rightarrow$ principal quantum number.

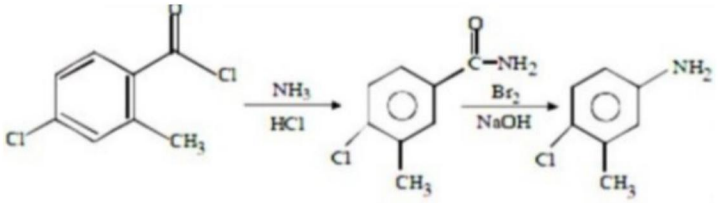
$\therefore$ statement I false statement II true.

36. $Na_2S + [Na_2Fe(CN)_5NO] \rightarrow Na_4[Fe(CN)_5NOS]$

Sodium nitro prusside — Violet coloured complex

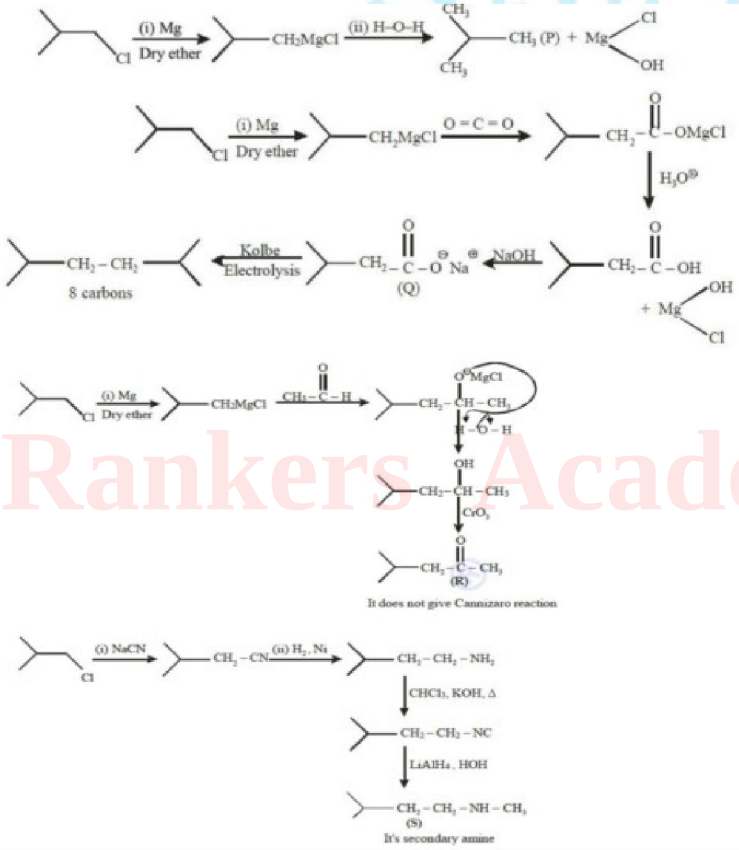
When Na_2S reacts with sodium nitro prusside forming violet coloured complex

37.



38. Reference NCERT (XII) page no 102, 103

39.



40. (I) As in Lyophobic colloids there is no interaction between dispersed phase and dispersion medium, special methods are used for preparation, simple mixing will not form colloid.

(II) Emulsions are liquid in liquid type colloids.

(III) Dissolving surfactant in a proper solvent will only form micelles at temperature above Kraft's temperature.

(IV) For Tyndall effect there must be a large difference in refractive index between dispersed phase and dispersion medium in order to have diffraction of light.

Hence ans (I) & (II) are correct.

41. Blue green colour is due to formation of $Cu(BO_2)_2$

$$CuSO_4 \xrightarrow{\Delta} CuO + SO_3$$

$$CuO_4 + B_2O_3 \rightarrow Cu(BO_2)_2$$

42. According to probability diagrams only for S orbital, wave function will have maximum at zero.

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43. $2C_2H_2(g)+5O_2(g)\rightarrow 4CO_2(g)+2H_2O(I)\Delta H^\circ/kJ=-2600kJ$ (i)

$C(s)+O_2(g)\rightarrow 2H_2O(I)\quad \Delta H^\circ/kJ=-390kJ$ (ii)

$2H_2(g)+O_2(g)\rightarrow 2H_2O(I)\quad \Delta H^\circ/kJ=-572kJ$ (iii)

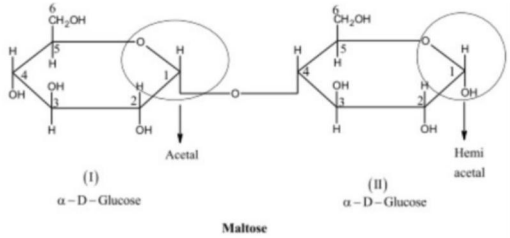
Δ_rH for $2C(s)+H_2(g)\rightarrow C_2H_2$ can be calculated by

$\Delta_rH=-(-1300)+2(-390)+\times(-572/2)$

$\Delta_rH=234$

44. (i) Ce, Pr, Nd, Tb and Dy shows +4 oxidation state
(ii) Sm, Eu, Yb $\rightarrow$ shows + 2 state
(iii) Lanthanides common oxidation sttae is +3
(iv) Eu shows only +2 or +3 states

45.

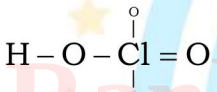


Maltose

46. Williamson's synthesis requires that the alkyl halide should be 1° and alkoxide may be 1°, 2° or 3°. Thus, the two ethers which cannot be prepared by Williamson's synthesis are :

$(C_6H_5)_2O,(CH_3)_3C-O-C(CH_3)_3$

47. $HClO_4$ structureis



No. of $Cl=0$ bonds $\rightarrow 3$

48. $\Delta T_{bcomplex}=2\Delta T_{bCaCl_2}$

$iK_b\times 0.1=2\times i\times K_b\times 0.05$

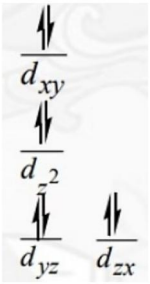
$CaCl_2\rightarrow Ca^{2+}+2Cl^-\qquad i_{CaCl_2}=3\qquad \Rightarrow i=3$

i.e. no. of Ions after dissociation of complex '3'

$[Cr(NH_3)_5Cl)Cl_2\rightarrow [Cr(NH_3)_5Cl]^-+2Cl^-\qquad \therefore x=5$

49. $\underset{\text{green}}{Ni^{2+}(aq)}+DMG\xrightarrow{AqNH_3}\underset{\substack{\text{Rosy Red voluminous ppt.}\\ \text{Hyd. of } Ni^{2+}:dsp^2}}{[Ni(DMG)_2]}\downarrow+2H^+$

Distribution of electrons in the d-orbitals of $Ni^{2+}(3d^8)$ in $[Ni(DMG)_2]:\overline{d_{x^2-y^2}}$



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50. $H_2 \rightarrow 2H^+ + 2e^-$

$M^{3+} 2e^- \rightarrow M^+$

---------- $E^0_{cell} = E^0_C - E^0_A = 0.2 - 0 = 0.2$

$H_2 + M^{3+} \rightarrow 2H \rightarrow 2H^+ + M^+$

$$E_{cell} = E^0_{cell} - \frac{0.0591}{2}\log\frac{[M^+][H^+]}{[M^{3+}]} = 0.2$$

$$0.1115 = 0.2 - \frac{0.059}{2}\log\frac{[M^+]}{[M^{3+}]}$$

$$-0.0885 = -0.0295\log\frac{[M^+]}{[M^{3+}]}$$

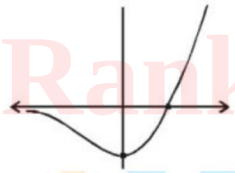
$$\log\frac{[M^+]}{[M^{3+}]} = 3 \Rightarrow \frac{[m^+]}{[M^{3+}]} = 10^3 = a = 3$$

MATHS

52. $$f'(x) = \frac{e^x.\left(x^3 - 3x^2 + \frac{9x}{2}\right)}{\left(x^2 + x + \frac{3}{2}\right)^2} = 0$$

$\Rightarrow x = 0$

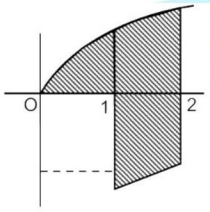
$\underset{x=0}{-\quad +}$



Function is $\uparrow$ in $(0, \infty)$

& function is $\downarrow$ in $(-\infty, 0)$

53. Required area $= \int_0^2 4\sqrt{x}dx + \frac{1}{2}(2023 + 2022)$



54. $1 + x + x^2 + x^3 = (1 + x) + x^2(1 + x) = (1 + x)(1 + x^2)$

$(1 + x + x^2 + x^3)^{11} = (1 + x)^{11}(1 + x^2)^{11}$

$(1 + {}^{11}C_1.x + {}^{11}C_2.x^2 + {}^{11}C_3.x^3 + {}^{11}C_4.x^4 + \ldots)$

$\times(1 + {}^{11}C_1.x^2 + {}^{11}C_2.x^4 + \ldots..)$

$\therefore$ The coefficient of x^4 is

${}^{11}C_2 + {}^{11}C_2 \quad {}^{11}C_1 + {}^{11}C_4 = 55 + 605 + 330 = 990$

55. Equation of the hyperbola is $xy - 5x - 8 = 0$.

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56. α and β lie on the chord $y=1$ intersecting the circle $x^2+y^2-8x-6y+17=0$

$\therefore x^2-8x+12=0 \Rightarrow x=2,6$

$\therefore \alpha=2+i, \beta=6+i$

57. Given, $(1+x^2)dy=y(x-y)dx$

Where, $y(0)=1, y(2\sqrt{2})=\beta$

$$dy=\left(\frac{yx-y^2}{1+x^2}\right)dx$$

$$\frac{dy}{dx}+y\left(\frac{-x}{1+x^2}\right)=\left(\frac{-1}{1+x^2}\right)y^2$$, divide by y^2 both sides and proceed

$$IF=e^{\int\frac{x}{1+x^2}dx}=e^{\frac{1}{2}\log(1+x^2)}$$

$$IF=\sqrt{1+x^2}$$

$$t\times\sqrt{1+x^2}=\int\frac{1}{x^2+1}\sqrt{x^2+1}dx$$

$$\frac{\sqrt{1+x^2}}{y}=\int\frac{1}{\sqrt{x^2+1}}dx$$

$$\frac{\sqrt{1+x^2}}{y}=\ln\left(x+\sqrt{x^2+1}\right)+\ln c$$

$$\frac{\sqrt{1+x^2}}{y}=\ln\left(c\{x+\sqrt{1+x^2}\right)$$

Put $x=0, y=1$

$$\frac{1}{1}=\ln(c(0+1))$$

$$\ln c=1\Rightarrow c=e$$

$$\frac{\sqrt{1+x^2}}{4}=\ln\left(e\left(x+\sqrt{1+x^2}\right)\right)$$

Put $x=2\sqrt{2}, y(2\sqrt{2})=\beta$

$$\frac{\sqrt{1+(2\sqrt{2})^2}}{\beta}=\ln\left(e\left(2\sqrt{2}+\sqrt{1+(2\sqrt{2})^2}\right)\right)$$

$$\frac{3}{\beta}=\ln\left(e\left(2\sqrt{2}+3\right)\right)$$

$$e(3+2\sqrt{2})=e^{3/\beta}$$

$$c(3+2\sqrt{2})=e^{3\beta^{-1}}$$

58. F(n) is not continuous at ±1.

59. Let $L=\int_{-\infty}^{\infty}\frac{dx}{1+x^2+x^4+\ldots x^{2m}}=2\int_0^{\infty}\frac{x^2-1}{x^{2m+2}-1}dx$

$$\lim_{m\to\infty}\frac{x^2-1}{x^{2m+2}-1}=1-x^2 \text{ if } 0\le x<1$$

$= 0$ for $x\ge1$

$$\Rightarrow L=2\int_0^1(1-x^2)dx=\frac{4}{3}$$

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60. Let $B=\int_0^{\pi}\frac{2x^2.\cos^2 x/2}{(1+\sin x)^2}dx$

$\therefore B-A=\int_0^{\pi}\frac{x^2(2\cos^2 x/2)}{(1+\sin x)^2}dx-\int_0^{\pi}\frac{x^2.\cos x}{(1+\sin x)^2}dx$

Using by parts,

$B-A=\left\{-\frac{x^2}{1+\sin x}\right\}_0^{\pi}+2\int_0^{\pi}\frac{x}{(1+\sin x)}dx$

$B-A=-\pi^2+2K$ …(i)

where, $K=\int_0^{\pi}\frac{xdx}{1+\sin x}$ $\left[\text{using}\int_0^{2a}f(x)dx=\begin{cases}0, f(2a-x)=-f(x)\\ 2\int_0^a f(x)dx, f(2a-x)=f(x)\end{cases}\right.$

$\Rightarrow 2K=2\pi\int_0^{\pi/2}\frac{dx}{1+\sin x}=2\pi\int_0^{\pi/2}\frac{dx}{1+\cos x}$

$==2\pi\int_0^{\pi/2}\frac{1}{2}\sec^2\frac{x}{2}dx=2\pi\left\{\tan\frac{x}{2}\right\}_0^{\pi/2}=2\pi$

$\therefore K=\pi$

Thus, $B-A-\pi^2+2K$

$\Rightarrow B-A-\pi+2\pi$

61. Let $\sec\theta=t$ and integrate to get

$F(\theta)=-\frac{1}{4}\ln\left(\frac{\sec\theta+1}{\sec\theta-1}\right)-\frac{1}{2}\tan^{-1}(\sec\theta)+\frac{1}{3\sec^3\theta}$

62. $\frac{6+2\log_3 x}{-5x}>0\ \&\ x>0\ \&\ x\neq\frac{1}{3}$

This gives $x\in\left(0,\frac{1}{27}\right)$ …(1)

$-1\le\log_{3x}\left(\frac{6+2\log_3 x}{-5x}\right)\le 1$

$3x\le\frac{6+2\log_3 x}{-5x}\le\frac{1}{3x}$

$15x^2+6+2\log_3 x\le 0$

$6+2\log_3 x+\frac{5}{3}\ge 0$

$x\in(0,c)$ …(2)

$x\ge 3^{-\frac{23}{6}}$ …(3)

From (1), (2), (3)

$x\in\left[3^{-\frac{23}{6}},c\right)$

$\alpha=3^{-\frac{23}{6}}$

$[\log_3\alpha]=\left[\frac{-23}{6}\right]=-4$

C is number smaller than $\frac{1}{27}$ but close to it.

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63. $\cot^{-1}\frac{5}{\sqrt{3}}+\cot^{-1}\frac{9}{\sqrt{3}}+\cot^{-1}\frac{15}{\sqrt{3}}+\cot^{-1}\frac{23}{\sqrt{3}}+\ldots\infty$

$=\tan^{-1}\left(\frac{\frac{2}{\sqrt{3}}-\frac{1}{\sqrt{3}}}{1+\frac{2}{\sqrt{3}}\times\frac{1}{\sqrt{3}}}\right)+\tan^{-1}\left(\frac{\frac{3}{\sqrt{3}}-\frac{2}{\sqrt{3}}}{1+\frac{3}{\sqrt{3}}\times\frac{2}{\sqrt{3}}}\right)+\ldots+\ldots+\tan^{-1}\left(\frac{\frac{n+1}{\sqrt{3}}-\frac{n}{\sqrt{3}}}{1+\frac{n+1}{\sqrt{3}}\times\frac{n}{\sqrt{3}}}\right)+\ldots.$

$\Rightarrow S_n=\tan^{-1}\frac{n+1}{\sqrt{3}}-\tan^{-1}\frac{1}{\sqrt{3}}$

$\lim_{n\to\infty}S_n=\frac{\pi}{2}-\frac{\pi}{6}=\frac{\pi}{3}$

64. Since $^{21}C_0+\ldots.+^{21}C_{10}+^{21}C_{11}+\ldots.+^{21}C_{21}=2^{21}$

$\Rightarrow$ but we have to select 10 objects and

$^{21}C_0+\ldots.+^{21}C_{10}=(^{21}C_0+\ldots.+^{21}C_{10})=2^{20}$

65. Let $C=\begin{bmatrix}1&2\\-1&-1\end{bmatrix},D=\begin{bmatrix}-1&-2\\1&1\end{bmatrix}$

$DC=\begin{bmatrix}1&2\\-1&-1\end{bmatrix}\begin{bmatrix}-1&-2\\1&1\end{bmatrix}=\begin{bmatrix}1&0\\0&1\end{bmatrix}=I$

$B=CAD$

$B^n\underbrace{(CAD)(CAD)(CAD)\ldots.(CAD)}_{n-\text{times}}$

$\Rightarrow B^n=CA^nD \quad \ldots.(1)$

$A^2=\begin{bmatrix}1&\frac{1}{51}\\0&1\end{bmatrix}\begin{bmatrix}1&\frac{1}{51}\\0&1\end{bmatrix}=\begin{bmatrix}1&\frac{2}{51}\\0&1\end{bmatrix}$

$A^3=\begin{bmatrix}1&\frac{3}{51}\\0&1\end{bmatrix}$

Similarly $A^n=\begin{bmatrix}1&\frac{n}{51}\\0&1\end{bmatrix}$

$B^n=\begin{bmatrix}1&2\\-1&-1\end{bmatrix}\begin{bmatrix}1&\frac{n}{51}\\0&1\end{bmatrix}\begin{bmatrix}-1&-2\\1&1\end{bmatrix}$

$=\begin{bmatrix}1&\frac{n}{51}+2\\-1&-\frac{5}{51}-1\end{bmatrix}\begin{bmatrix}-1&-2\\1&1\end{bmatrix}=\begin{bmatrix}\frac{n}{51}+1&\frac{n}{51}\\-\frac{n}{51}&1-\frac{n}{51}\end{bmatrix}$

$\sum_{n=1}^{50}B^n=\begin{bmatrix}25+50&25\\-25&-25+50\end{bmatrix}=\begin{bmatrix}75&25\\-25&25\end{bmatrix}$

Sum of the elements = 100

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66. $A = \{1,2,3,4,5,6,7\}$

$R = \{(x,y) \in A \times A; x + y = 7\}$

$x + y = 7$

$y = 7 - x$

$E = \{(1,6),(2,5),(3,4),(4,3),(5,2),(6,1)\}$

$(a,b) \in R \Rightarrow (b,a) \in R$

$\Rightarrow$ Relation is symmetric.

67. Given mean $(\bar{x}) = \frac{9}{2}$

$$\bar{x}_{new} = \frac{12 \times \frac{9}{2} + 7 + 14 - 9 - 10}{12} = \frac{14}{3} \quad \ldots(1)$$

Given, $\sigma^2 = 4$

$$\sigma^2 = \frac{\sum x_i^2}{12} - \left(\frac{9}{2}\right)^2$$

$$4 = \frac{\sum x_i^2}{12} - \frac{81}{4}$$

$$\frac{\sum x_i^2}{12} - \frac{97}{4}$$

$$\sum x_i^2 = 291$$

Now,

$$\sum \left(x_i^2\right)_{new} = 291 - 9^2 - 10^2 + 7^2 + 14^2 = 355$$

$$\therefore \sigma_{new}^2 = \frac{\sum \left(x_i^2\right)_{new}}{12} - (\bar{x}_{new})^2$$

$$\sigma_{new}^2 = \frac{355}{12} - \left(\frac{14}{3}\right)^2 = \frac{281}{36} \text{ (from eq. (i))}$$

68. Probability of any event cannot be greater than 1

69. Image of point (3, 5) in $x - y + 1 = 0$ is

$$\frac{x-3}{1} = \frac{y-5}{-1} = \frac{-2(3-5+1)}{1^2+1^2}$$

$$\Rightarrow \frac{x-3}{1} = \frac{y-5}{-1} = 1$$

$\Rightarrow x = 4, y = 4$

From the given options, option(a) satisfies it.

70. $S = 44\frac{1}{2}$

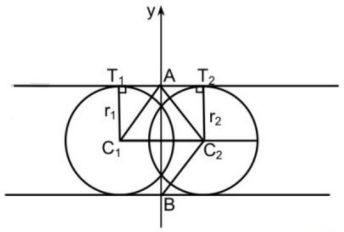
71. $L_1 : \frac{x-8}{8} = \frac{y-0}{-6} = \frac{z-0}{0}$

$L_2 : \frac{x}{0} = \frac{y+8}{8} = \frac{z}{-5}$

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$$d = \frac{\begin{vmatrix} 8 & 8 & 0 \\ 8 & -6 & 0 \\ 0 & 8 & -5 \end{vmatrix}}{\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 8 & -6 & 0 \\ 0 & 8 & -5 \end{vmatrix}} = \frac{280}{\sqrt{1649}}$$

72.



$AT_1 = AT_2$

$C_1A^2 = C_1T_1^2 + AT_1^2$

$C_2B^2 = C_2A^2 = C_2T_2^2 + AT_2^2$

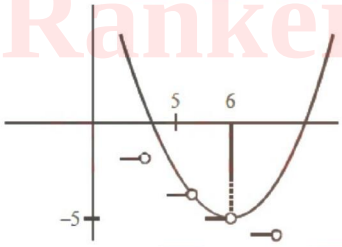
$C_1A^2 - C_2B^2 = r_1^2 - r_2^2 = \frac{91}{4}$

73. $x^2 - 12x + [x] + 31 = 0$

$x^2 - 12x + 31 = -[x]$

$(x-6)^2 - 5 = -[x]$

By graph



Zero point of intersection, m = 0

$x^2 - 5|x+2| - 4 = 0$

Case-I: $x < -2$

$x^2 + 5x + 6 = 0$

$x = -3, -2$(rejected)

Case-II; $x \geq -2$

$x^2 - 5x - 14 = 0$

$x = 7, -2$

Number of solution (n) = 3

So $m^2 + mn + n^2 = 9$

74. Since, m, n are indentical and independent

$$2S = \sum_{m=0}^{\infty}\sum_{n=0}^{\infty} \frac{m+n+mn}{2^m + 2^n}\left[\frac{1}{2^m} + \frac{1}{2^n}\right] = \sum_{m=0}^{\infty}\sum_{n=0}^{\infty}(m+n+mn)\frac{1}{2^m}\cdot\frac{1}{2^n}$$

$$= \sum m.\left(\frac{1}{2}\right)^m \sum_0^{\infty}\left(\frac{1}{2}\right)^n + \sum_0^{\infty}\left(\frac{1}{2}\right)^m \sum_0^{\infty} n\left(\frac{1}{2}\right)^n + \sum_0^{\infty} m\left(\frac{1}{2}\right)^m \sum_0^{\infty} n\left(\frac{1}{2}\right)^n$$

$= 4 + 4 + 2.2 = 12 \Rightarrow S = 6$

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75. $2(\vec{a}\times\vec{c})+3(\vec{b}\times\vec{c})=\vec{0}\Rightarrow(2\vec{a}+3\vec{b})\times\vec{c}=\vec{0}$

So $\vec{c}\,\|\,(2\vec{a}+3\vec{b})=(7,-13,19)$. Let $\vec{c}=t(7,-13,19)$

$\vec{a}-\vec{b}=(1,-4,2)$.

$(\vec{a}-\vec{b}).\vec{c}=t(7+52+38)=97t=-97\Rightarrow t=-1$

$\vec{c}=(-7,13,-19)$

$$\vec{c}\times\hat{k}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\-7&13&-19\\0&0&1\end{vmatrix}=(13,7,0)$$

$|\vec{c}\times\hat{k}|^2=169+49=218.$

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